How IT Failure Creates Value: Recoverable Disruption in Continuous Feature Innovation[*]

Ankur Jaiswal[‡] Hüseyin Tanriverdi

McCombs School of Business, The University of Texas at Austin

Research in Progress

Preliminary Draft: Please Do Not Cite or Circulate

October 22, 2026

Click here for the latest version.

Abstract

Digital platforms evolve through continuous feature releases into systems knowledge workers already depend on, making innovation and reliability inseparable. Reliability research treats resulting failures as costs offsetting innovation, but this view is incomplete. Using Real User Monitoring traces of 34 million actions from an InsurTech CRM serving insurance agencies, where agent performance is measured by sales conversion, we estimate a causal mediation system in which feature use and IT failure are separately endogenous, each with its own shift-share instrument. Feature use raises performance. It also raises IT failure. Failure in turn raises performance, and this indirect pathway accounts for 22% of the total effect, so part of the business value of feature innovation is created through the failures it generates. Decomposing failure across five categories of knowledge work shows why. Recoverable internal failures are followed by retry and fast resolution and contribute positively. Customer-facing failures are followed by abandonment and contribute negatively. Recoverability determines whether IT Failure creates or destroys value.

Keywords: *IT failure; business value; recoverability; coping; feature innovation; knowledge work; sales performance; CRM; causal mediation*

[*]I am especially grateful to all the members of my dissertation committee, Sirkka L. Jarvenpaa, Ashish Agarwal, Raghunath Singh Rao and Ram Ranganathan, for their detailed feedback and continuous guidance throughout the various stages of this project. I also thank Ayush Kanodia for his help at many points in this project. I am very grateful to the anonymous Insurance CRM platform for access to the proprietary data analyzed here. They gave generously of their time in explaining how the system is structured and data are recorded, and how the work it supports is carried out in practice. That guidance continues to shape this study, the essays of my dissertation, and the future stream of work that I am building upon.

[‡]Email: ankur_jaiswal@utexas.edu

Introduction

Digital innovation increasingly occurs not through episodic replacement of information systems, but through the continuous introduction of new features into systems already embedded in organizational work (Nambisan et al., 2017; Yoo et al., 2010). This shift changes an important feature of technology-enabled work. Rather than separating innovation from stable operation, digital platforms introduce new capabilities while users continue to depend on the same systems to perform ongoing tasks. Continuous feature innovation can therefore expand what a digital system enables workers to do while simultaneously altering workflows, creating new interdependencies, and introducing opportunities for IT failure.

This creates a fundamental tension between innovation and reliability. Research on digital innovation emphasizes how new technological capabilities can enable new or improved forms of work. Research on IT reliability, in contrast, generally treats failures as disruptions that interfere with task execution and diminish the value organizations obtain from technology (Addas & Pinsonneault, 2018; Butler & Gray, 2006). At the organizational level, IT failures can destroy market value and impose substantial organizational costs (Bharadwaj et al., 2009; Triche & Walden, 2018). Taken together, these perspectives imply a tradeoff: new features may create value through improved capabilities, but the failures accompanying those features may offset some of those gains.

Yet this expectation overlooks an alternative possibility. Failure does more than interrupt work. It can reveal information that uninterrupted operation does not, disrupt established routines, and prompt users to reconsider how a task is being performed. Research on learning from failure suggests that relatively small failures can generate learning when individuals can respond quickly and when the underlying domain is sufficiently familiar to permit adaptation (Sitkin, 1992). In continuously evolving digital systems, some failures may therefore induce workers to re-engage with a task, retry unsuccessful actions, and adapt their behavior. If so, the failures generated by digital innovation may sometimes transmit rather than dissipate its performance benefits.

We argue that a critical distinction is the recoverability of the work disrupted by IT failure. When failure interrupts internal work that remains available to the worker—for example, entering, updating, or retrieving information—the worker can return to the task, retry the unsuccessful action, and continue. Such failures create disruption, but they do not necessarily eliminate the underlying opportunity to complete the work. By contrast, when failure occurs during externally paced work, such as communicating with a waiting customer, the interrupted interaction may be difficult or impossible to recover. The consequences of IT failure may therefore depend not simply on whether technology fails, but on whether the disrupted work permits recovery. Recoverable failures can create opportunities for re-engagement and adaptation; failures that compromise the underlying task opportunity are more likely to remain costly.

This perspective raises a broader question for research on digital innovation: How do IT failures transmit the performance effects of continuous feature innovation, and when can the disruptions created by those failures enhance rather than diminish performance? Answering this question requires observing more than whether new technology is available or adopted. It requires tracing the pathway from the introduction and use of new features, through the failures workers encounter, to their subsequent behavior and ultimately to measurable performance.

Literature Review

IT Failure and Organizational Value

Research on IT failure has concentrated at the organizational level. Bharadwaj et al. (2009) apply event study methodology to publicly reported IT failures and find reductions in firm market value. Triche and Walden (2018) examine 214 IT failures over eight years and find that a firm's choice of impression management strategy affects its market value, with assertive strategies performing better for implementation failures and defensive strategies performing better for operational failures. Davis et al. (1992) address the interpretation of failure episodes, developing a diagnostic framework on the premise that relevant data must first be identified, collected and organized before the perspectives, actions and events producing a failure can be reconstructed.

Banker et al. (2021) study technical debt accumulated in the CRM systems of 26 firms over eleven years. They report that shortcuts taken during deployment reduce system reliability and can cause disruptions that impair firm performance, and that the same shortcuts allow faster deployment and idiosyncratic functionality that can raise it. They also find that firms in industries with higher clockspeed and higher competitive threat accumulate more technical debt.

Najjar et al. (2022) examine what follows a failure. Applying service-dominant logic to IS incident resolution, they identify responsive handling, supportive communicating and effective resolving as components of a satisfying recovery, and find that user recovery satisfaction depends both on the speed and completeness of the fix and on the effort and fairness conveyed during it. Recovery in their account is a property of the provider's response. The present study uses the term for a property of the interrupted work. In each of these studies the outcome is measured at the level of the firm or the organization, and the failure is observed through public disclosure or through incident records.

IT Interruptions and Individual Performance

A separate body of work examines interruptions at the individual level. Addas and Pinsonneault (2015) develop a taxonomy classifying IT interruptions by the relevance and structure of their

content, validated through log diaries and interviews with professional workers. They report that some IT interruptions have positive effects on individual performance, others have negative effects, and some have both, and they attribute the inconsistent findings of earlier research to the absence of such a distinction. Addas and Pinsonneault (2018) apply this to e-mail interruptions and identify conditions under which a benefit arises.

Weinert et al. (2022) examine repeated exposure. In a laboratory experiment with 100 subjects analysed through a multivariate latent growth model, they find that users may habituate to repeated IT interruption or become sensitized to it, and that the effects of arousal and exhaustion on task performance vary over time. An interruption in these studies is an event that captures attention while the worker is engaged in something else. A failure is the technology not performing when the worker invokes it. The two differ in what the worker was doing when the event occurred and in whether the intended task remains available afterward.

Worker Response and Adaptation

Research on user adaptation examines how workers respond to IT events. Beaudry and Pinsonneault (2005) define user adaptation as the cognitive and behavioural efforts users perform to cope with significant IT events, and derive four adaptation strategies from two appraisals. Primary appraisal concerns the expected consequences of the event. Secondary appraisal concerns the user's assessment of control over the situation. In their model, performance gains arise in the strategies associated with high control.

Elie-Dit-Cosaque and Straub (2011) study the antecedents of perceived behavioural control over IT and find that autonomy, managerial support, reduced work overload and perceived innovativeness with IT together raise it. Pirkkalainen et al. (2019) distinguish deliberate proactive coping from instinctive reactive coping in a survey of 846 employees and find that the two operate differently on IT-enabled productivity. Mattern et al. (2024) combine interviews with heart rate variability data from fifteen high-performing IT professionals and theorize coping portfolios, in which distinct combinations of coping behaviour correspond to distinct strain levels. Rinta-Kahila et al. (2023) trace the opposite condition, showing through a case study of cognitive automation that skill erosion leaves workers unable to respond when the automation fails. These studies measure response through survey instruments, laboratory manipulation, or intensive observation of small samples.

Continuous Feature Innovation and the Origins of Failure

Digital innovation research characterizes digital technologies by their recombinability and continuing evolution (Nambisan et al., 2017; Yoo et al., 2010). Feature-level perspectives examine

how users engage particular capabilities within systems already in use (Karahanna et al., 2019). Tiwana and Safadi (2024) construct a trace dataset from 190 million lines of evolving code across 1,354 systems spanning 25 years and find that systems atrophy as they age, that atrophy increases bugginess and disengages developers, and that increments in modularity slow atrophy but lose potency with age.

Vieira da Vieira da Cunha et al. (2015) study a CRM system through a fifteen-month ethnography of a sales department. They distinguish settings in which work and its electronic representation are tightly coupled, where the system records work automatically, from settings in which the two are loosely coupled, where employees report their own work into the system. Complexity research treats such environments as adaptive systems in which agents adjust through local interaction and the information that interaction returns (Benbya et al., 2020; Nan & Tanriverdi, 2017).

Table 1 codes twenty-five studies from these literatures on a common schema of construct, unit of analysis, mechanism, key assumption, and the distance between the measure used and the phenomenon of interest. Seven studies draw on trace or archival data and measure outcomes at the level of the firm or the system. Seven rely on self-report. Eight are qualitative or conceptual. Two use laboratory or physiological measurement. One combines field data with an objective performance outcome.

[Insert Table 1 about here]

Consequences and behaviour have therefore been measured in different places. Studies that observe performance consequences do not observe what workers do. Studies that observe what workers do rely on recall or on settings constructed for the purpose. Whether the performance consequence of a failure depends on the worker's response has not been tested directly.

This study observes the release of new features, the failures those releases generate, the behaviour of workers in the moments following each failure, and the sales outcomes that follow, at a single unit of analysis in one setting.

Methodology

Empirical Setting and Sample

We investigate this question using proprietary data from a leading InsurTech SaaS CRM provider whose platform serves more than 1,000 small and medium independent insurance agencies across the United States.[1] The empirical setting allows us to observe the relationship between contin-

[1] Access to the data is governed by a signed non-disclosure agreement with the business provider. All regression analyses are estimated on the raw, unscaled data. The platform collectively manages premiums exceeding $8 billion across diverse property and casualty portfolios; variables that could reveal commercially sensitive scale or operational details are reported after applying confidential scaling and standardization.

uous digital innovation and work at unusually fine granularity. We combine structured pipeline records[2] with Real User Monitoring trace data[3] capturing more than 34 million user actions across approximately half a million user sessions and more than four million task initiations by users from January to July 2024. The platform spans over 200 workflow modules and 145,000 interaction configurations, whose interdependencies constitute the compositional complexity of the workflows agents navigate. These data allow us to observe agents' use of newly released features[4], IT failures encountered while performing work, post-failure behavior, and subsequent sales conversion performance at the agent-week level. Variable definitions and the composition of the failure measure across task categories and by cause are reported in Tables 2 to 4. The sample is constructed by applying inclusion criteria at the agency, agent and task levels, so that observations reflect active, observable customer acquisition activity on the platform. Agencies are included if they maintain an active acquisition pipeline, generate a minimum volume of monthly business during the study window, and have at least a minimum number of licensed agents operating on the platform; the thresholds are withheld under the non-disclosure agreement. Agents are included where trace data are available for the study window, so that feature use, failure and post-failure behaviour are observed on the same agent-weeks as performance. The estimation sample for the main results is the core sample; measures defined on only a subset of agent-weeks are estimated on correspondingly fewer observations, as each table reports.

[Insert Tables 2 to 4 about here]

Econometric Model

Decomposing the total effect of feature use requires sequential ignorability (Imai et al., 2010), and both of its conditions fail here for structurally independent reasons: feature use is a behavioural choice correlated with unobserved time-varying agent-level factors, while IT failure reflects platform-side conditions driven by distinct forces. The agent-level factors at issue are

[2]A dealcard is a centralized record serving as the core unit of analysis for agents and CRM system developers, functioning as a data repository for each sales opportunity — capturing customer details, product type, pricing, and process updates as the deal progresses through the pipeline. The acquisition pipeline averages 312,000 dealcard records over the study period. A dealcard is equivalently a sales opportunity (Hartmann et al., 2025).

[3]Trace data consist of timestamped records of user actions within the system, capturing behaviours such as logging in or out, page views, and button clicks. Over the study window the trace comprises over 34 million action events across approximately half a million user sessions and over 4 million task initiations, each a single invocation of a platform feature to deliver a defined functional capability. From a developer's perspective, it also captures computational resources used, bugs, frictions, and modular environments affecting deployment of features and modular integrations. These data are collected unobtrusively as by-products of system usage, providing a detailed log of activities as users engage with the platform. The literature describes trace data as "residues left behind by multiple types of behavior" (Lindberg, 2020) and as "logs of employee behavior," characterised by natural occurrence rather than active production for research purposes (Grover et al., 2020).

[4]The CRM features examined were identified through systematic sampling in consultation with the platform, selecting features most relevant to the sales agent; specific counts and release dates are withheld under the NDA.

unobserved time-varying skill, effort, opportunity quality and workload, any of which may raise conversion while also raising how intensively an agent takes up newly released features. That independence forecloses single-instrument identification. Dippel et al. (2020) recover mediation effects with one instrument only by assuming that treatment and mediator share a common unobserved confounder, which is precisely what fails here; when the endogeneity sources differ, each endogenous variable requires its own instrument (Celli, 2022; Frölich & Huber, 2017). We therefore estimate a linear structural system with a dedicated instrument for each.

$$IT\ Failure_{it} = \beta_1\, New\ CRM\ Feature\ Use_{it} + \boldsymbol{\gamma}_1'\mathbf{X}_{it} + \alpha_i + \tau_t + v_{it} \tag{1}$$

$$Performance_{it} = \varphi_1\, New\ CRM\ Feature\ Use_{it} + \boldsymbol{\gamma}_2'\mathbf{X}_{it} + \alpha_i + \tau_t + u_{it} \tag{2}$$

$$Performance_{it} = \delta_1\, New\ CRM\ Feature\ Use_{it} + \delta_2\, IT\ Failure_{it} + \boldsymbol{\gamma}_3'\mathbf{X}_{it} + \alpha_i + \tau_t + \varepsilon_{it} \tag{3}$$

New CRM Feature Use$_{it}$ and *IT Failure*$_{it}$ are the endogenous variables, and each is instrumented separately. Equation (1) is the mediator equation, reported in Table 5. Equation (2) is the reduced form, in which performance is regressed on feature use with the mediator omitted, so that φ_1 carries the direct and indirect pathways together and estimates the total effect in one step; it is reported in column 2 of Table 6. Equation (3) adds the mediator and separates the two. *Performance*$_{it}$ is agent i's sales performance in week t, operationalised as conversion, the rate at which active later-stage deals resolve as successful purchases in that agent-week; the exhibits report it under that name. *IT Failure*$_{it}$ is the share of that agent-week's session time spent on tasks that do not complete. δ_1 is the direct pathway from feature use to performance, $\beta_1\delta_2$ is the indirect pathway running through failure, and $\delta_1 + \beta_1\delta_2$ is the total effect, which φ_1 estimates directly.

[Insert Table 5 about here]

[Insert Table 6 about here]

$\mathbf{X}_{it}$ is the same control vector in all three equations and contains *New GenAI Use*, *Incidence of Failure*, *Feature Breadth*, *Platform Time*, *Product Complicatedness* and *Customer Complicatedness* at both the qualification and the conversion stage, *Process activity burden*, *Bundling intensity*, *Portfolio Value*, *Portfolio Age* and *Portfolio Size*. α_i and τ_t are agent and week fixed effects. The controls are not constrained to take the same coefficients across the three equations, so each equation carries its own $\boldsymbol{\gamma}$.

Those fixed effects, together with the controls in $\mathbf{X}$, address only part of the problem. Agent fixed effects absorb what is time-invariant about an agent, such as persistent skill, and week fixed effects absorb what is common to all agents in a week, such as seasonal demand. Neither absorbs a shock specific to one agent in one week, nor the possibility that feature use responds within the

week to the same conditions that produce failure, so that part of an apparent mediation effect would reflect reverse causality.

A design that randomly assigned each agent's feature use and failure exposure every week would address both problems: randomising feature use independently of unobserved skill would make skill orthogonal to it, and randomising failure independently of feature use would break the path running from performance back to the mediator. Neither can be assigned experimentally in the field, so we pursue a quasi-experimental design in which each endogenous variable is instrumented separately, drawing on shocks unrelated to any individual agent's performance. This requires a level of variation below the agent: shares and shocks are defined over the platform's CRM workflow modules, the modular components through which agents complete their work, each separately identifiable in the trace data by the module associated with every user action.

Both equations are estimated by two-stage least squares (2SLS) with standard errors clustered at the agent level; full estimates for all three appear in Table 5 and Table 6. Because the ACME is a product of coefficients, its sampling distribution is skewed, and the Sobel (1982) normal approximation yields low power and unbalanced confidence limits (Fritz & MacKinnon, 2007; MacKinnon et al., 2004). Inference therefore relies on a cluster bootstrap resampling agents, re-estimating the mediator and outcome equations within each replicate so that the correlation between the paths propagates; bias-corrected 95% intervals are the inferential criterion.

Identification Strategy

Both endogenous variables, *New CRM Feature Use*$_{it}$ and *IT Failure*$_{it}$, are instrumented using the exogenous-shifts shift-share framework of Borusyak et al. (2022, 2025). Each instrument takes the form

$$Z_{it} = \sum_{m} s_{im,t-1}\, g_{mt} \tag{4}$$

where m indexes the platform's CRM workflow modules, $s_{im,t-1}$ is agent i's lagged share of workflow engagement attributable to module m, and g_{mt} is a module-week shock determined at the platform level. Shares are built from the prior week's click activity and normalised so that $\sum_m s_{im,t-1} = 1$ within each active agent-week, which makes them predetermined relative to the contemporaneous shock.[5] Identification under this design also rests on two conditions on the shocks

[5]The trace data record every user action together with the workflow module in which it occurs, so engagement can be aggregated to the agent-module-week level. Shares are formed from click counts in the preceding week and sum to one within each active agent-week.

themselves: dispersion[6] and independence[7] Both conditions are maintained in the present analysis, pending shock-level diagnostics.

The share construction rests on the structure of the work itself. Advancing a deal requires the joint completion of tasks spread across several interdependent workflow modules, so no single module carries a deal on its own: an O-ring production structure (Demirer et al., 2026; Gans & Goldfarb, 2026; Kremer, 1993). An agent's exposure to a platform-level disturbance is therefore governed by how that agent's work is distributed across modules, which is what the click-based shares measure. Identification is attributed entirely to quasi-random variation in the shocks. Under the exogenous-shifts route the orthogonality condition applies to the shift rather than to the shares, which is what permits the exposure shares themselves to be endogenous.

Instrument for New CRM Feature Use

$$Z_{it}^{\text{feature}} = \sum_{m} s_{im,t-1}\, g_{mt}^{\text{feature}}, \qquad g_{mt}^{\text{feature}} = \sum_{w} \mathbf{1}\{r_{wm} \le t\} \tag{5}$$

The shock g_{mt}^{feature} is the cumulative stock of platform-controlled CRM feature releases active in module m by week t, where r_{wm} is the week in which release wave w takes effect in module m.[8] The platform introduced new features through successive, staggered release waves, and the cumulative shock accumulates active release exposure across those waves as each takes effect week by week. Release timing and scope are set by the platform's product-development decisions at the system level and vary across workflow modules. No individual agent can alter when a release ships or which module it reaches: releases are deployed platform-wide rather than directed at any agent, and a single agent's engagement is a negligible share of the activity in any module. The shift is therefore plausibly independent of any individual agent's conversion outcomes within the same week.

[6] If the shocks are concentrated in a small number of module-weeks rather than spread across many, the instrument effectively collapses to a handful of large shocks, and the asymptotic properties that justify treating the shift-share estimator as consistent no longer apply (Borusyak et al., 2022).

[7] If the shocks are correlated with one another, the design carries less independent identifying variation than the number of module-weeks suggests (Adão et al., 2019), and a shock correlated with an omitted, module-wide confound could present as exogenous variation in *New CRM Feature Use* or in *IT Failure*. The exogenous-shifts route avoids defending the exogeneity of agents' module exposure shares directly, a requirement Goldsmith-Pinkham et al. (2020) show is demanding and one we consider implausible here, since agents' historical module reliance reflects their own workflow patterns.

[8] The platform introduced new CRM features in staggered release waves across the study window, each wave expanding capability in a subset of workflow modules. The shock accumulates the waves that have taken effect in a module by week t, so it is a dated treatment-intensity measure rather than a post-release indicator. Release dates and feature identities are withheld under the non-disclosure agreement.

Instrument for IT Failure

$$Z_{it}^{\text{failure}} = \sum_{m} s_{im,t-1}\, g_{m,t,-a(i)}^{\text{failure}}, \qquad g_{m,t,-a(i)}^{\text{failure}} = \sum_{j \notin a(i)} \sum_{v \in V_{jmt}} e_v \tag{6}$$

The shock is the module's weekly error load, defined as the total number of runtime errors a CRM workflow module raises across every action executed against it on the platform in week t by agents outside the focal agency. V_{jmt} is the set of actions agent j executes against module m in week t, e_v is the number of errors raised while action v executes, and $a(i)$ is the focal agent's agency; the outer sum is the population of agents the shock is built from, and the condition on it is what removes the focal agency. Errors enter the shock through the same log that defines failure in the measure: a task joins F_{isk} when at least one error is logged against it, and e_v counts those same logged errors at the level of the action that raised them.[9]

A module is a subsystem of the platform whose internal linkages are exercised each time its interface is assembled. Near-decomposable systems are characterised by the frequency of intra-component linkage relative to intercomponent linkage (Simon, 1962). An error is a coded dependency that does not resolve. Whereas the module's dependency structure characterises the linkages it must complete, its error load characterises that structure as evaluated at the code state then deployed.

Variation in this quantity across weeks originates in the platform's development cycle. A code change introduces a defect, the platform's developers monitor error incidence continuously and remediate detected defects, and the module returns to its maintained baseline. The series is accordingly a sequence of transient deviations generated by release and remediation decisions, which the platform's development team takes on its own schedule in response to platform-wide telemetry rather than to the work of any individual agent.

The series carries no individual agent's behaviour. Each error requires an action to raise it, but the shock aggregates all actions taken against the module by agents outside the focal agency, so that aggregation recovers the module's propensity to fail under its current code state rather than the conduct of any single agent. The focal agency is excluded in full, which removes the agents whose portfolio composition and demand conditions most closely resemble those of the focal agent. Two features of the construction further restrict the variation entering the instrument. Exposure shares

[9]The measure is the real-user-monitoring error count for a CRM workflow module, defined as the number of runtime errors raised while the module's interface executes one action, summed over the platform's actions against that module in the week. These comprise unhandled exceptions and errors written to the browser console. Systems research characterises a web page by the content fetched in rendering it (Butkiewicz et al., 2011) and treats a page load as a structure of linked activities whose dependencies can be recovered and traced (Wang et al., 2013), so a dependency that does not resolve is recorded as an error on the same trace. Real-user monitoring is the established means of observing a deployed system as it actually operates, in place of synthetic conditions (Cito et al., 2015), which is what allows the series to be read as realised subsystem activity. A task initiation is classified as failed when at least one error is logged against it.

are lagged and sum to one within each agent-week, so an increase in error incidence common to all modules in a given week enters the instrument as a constant and is absorbed by the week fixed effects; only relative variation across modules is retained. Agent fixed effects absorb persistent differences in the modules through which an agent works. The shift is therefore plausibly independent of any individual agent's conversion outcomes within the same week.

Under the exogenous-shifts route the orthogonality condition applies to the shift rather than to the shares, so the exposure shares themselves require no exogeneity defence (Borusyak et al., 2022). Underidentification and weak-instrument diagnostics for both first stages are reported in Table 5 and Table 6.

Exclusion Restrictions

Z_{it}^{feature} is required to affect sales performance only through $\textit{New CRM Feature Use}_{it}$, the agent's novelty-weighted intensity of engagement with newly released CRM features. Platform-controlled deployments expand the functionality available within a workflow module but do not themselves generate sales outcomes; performance is affected only insofar as agents engage with the released features, which is what $\textit{New CRM Feature Use}_{it}$ measures. Agent and week fixed effects absorb time-invariant differences in workflow exposure and common temporal shocks that may coincide with release timing.

Z_{it}^{failure} is required to affect performance only through $\textit{IT Failure}_{it}$, the focal agent's IT failure rate. Error incidence among agents in other agencies reflects platform-level infrastructure conditions in shared workflow modules; these conditions propagate to the focal agent through shared system exposure, raising the failure probability in that agent's own task initiations while having no direct bearing on that agent's sales outcomes. The leave-one-agency-out construction removes within-agency channels, such as correlated performance shocks or resource-sharing arrangements, through which neighbouring error incidence could reach the focal agent's sales outcomes by a route other than $\textit{IT Failure}_{it}$. Both restrictions are maintained assumptions rather than testable implications of the design.

Results

Column 2 of Table 5 and columns 4 and 5 of Table 6 are estimated by two-stage least squares, using the instruments set out in the identification strategy; the remaining columns are ordinary least squares (OLS). The causal effects reported below come from those specifications.

The Causal Mediation System

Greater use of newly released CRM features causally increases IT failure. The OLS association is negative, −0.022 (p<0.05, column 1 of Table 5), identified across agents who have chosen different levels of *New CRM Feature Use*. The causal effect is 0.248 (p<0.01, column 2), with Kleibergen–Paap rk LM 62.98 and Cragg-Donald Wald F 159.40. The consequences of that increase for performance are estimated in the outcome equation.

Under OLS there is no mediation. Column 2 of Table 6 is the reduced form, where the coefficient on *New CRM Feature Use* is 0.018 (p<0.05) and is the OLS total effect. Adding *IT Failure* in column 3 leaves that coefficient at 0.019 and puts the mediator at 0.023 (p<0.05), so under OLS the mediator accounts for almost none of the association with *Conversion*.

The causal effects are substantially larger than their OLS counterparts and carry the same signs. In column 4 of Table 6, where *New CRM Feature Use* alone is treated as endogenous, the causal effect of *New CRM Feature Use* on *Conversion* is 0.291 (p<0.01) and the coefficient on *IT Failure* is 0.034 (p<0.01). In column 5, where *New CRM Feature Use* and *IT Failure* are both treated as endogenous, the causal effects are 0.235 (p<0.01) and 0.266 (p<0.05) respectively. First stages are strong in both columns, with Kleibergen–Paap rk LM of 63.53 and 79.97 and Cragg-Donald Wald F of 162.93 and 65.94. The positive causal effect of *IT Failure* on *Conversion* is the result a reliability perspective does not predict.

Both instruments are strongly relevant. The Sanderson–Windmeijer F is 68.30 in the single-endogenous system of column 2 of Table 5, and 64.38 for *New CRM Feature Use* and 88.22 for *IT Failure* in the two-endogenous system of column 5 of Table 6 (Table 7). The coefficient on Z_{it}^{feature} is negative because the novelty weight a_{mt} falls as an innovation diffuses and adoption spreads, so modules with more cumulative releases carry more widely adopted features and a lower novelty-weighted index; the direction does not bear on identification, which turns on how strongly the instrument moves the variable. Both instruments enter each first stage, as system estimation requires, and the Sanderson–Windmeijer statistics confirm that each endogenous variable is separately identified.

[Insert Table 7 about here]

The indirect pathway accounts for approximately 22 percent of the total effect of feature use on conversion. Table 8 reports the causal mediation results from column 2 of Table 5 and column 5 of Table 6: the a-path is 0.248 (p<0.01) and the b-path 0.266 (p<0.05), giving an average causal mediation effect of 0.066 (p<0.05), an average direct effect of 0.235 (p<0.01) and an average total effect of 0.300 (p<0.01). Bias-corrected bootstrap intervals exclude zero for all three. The average total effect is estimated on the two-stage least squares system and is therefore not directly comparable with the OLS total effect of 0.018 in column 2 of Table 6.

[Insert Table 8 about here]

Where Failure Arises

Failure is concentrated in communicating and recording work. Failure occupies 2.67% of session time. Of that total, communicating accounts for 47.1% of incidence and 41.3% of failure time, and recording for 35.0 and 38.2% (columns 2 and 3 of Table 3); integrating follows at 11.5 and 15.4%, and coordinating and curating together account for under 7% on either basis.

The two largest categories differ in what causes their failures. Across all failure time, 41.9% is system-caused and 58.2% user-caused (columns 2 and 3 of Table 4). Communicating failures are predominantly system-caused, at 30.9 of their 41.3 percentage points, while recording failures are almost entirely user-caused, at 37.7 of 38.2.

The aggregate failure effect arises from internal work and is partly offset by customer-facing work. Figure 1 reports the decomposition of the aggregate effect of 0.062 across the five categories: the contribution of recording is 0.064 ($p<0.05$) and that of integrating 0.037 ($p<0.10$), while the contribution of communicating is -0.051 ($p<0.05$), and coordinating and curating are not distinguishable from zero. The decomposition is descriptive rather than causal, so we use the trace data to examine behaviour following each form of failure.

[Insert Figure 1 about here]

How Workers Respond

Workers persist after failures in internal work and redirect their effort after failures in customer-facing work. Table 9 reports six regressions of standardized post-failure behaviours on the two failure blocs. After internal-facing failure, *Page retry* rises by 0.069 ($p<0.01$, column 1), *Short reload* by 0.106 ($p<0.01$, column 2), and *Recovery time* falls by 0.319 ($p<0.01$, column 3). After customer-facing failure, *Page retry* and *Short reload* are unchanged (columns 1 and 2), *Recovery time* falls by less, 0.132 against 0.319 (column 3), and *Reroute rate* rises by 0.659 ($p<0.01$, column 4). *Abandonment* and *Workflow deviation* rise after both forms of failure, at 0.916 against 0.549 in column 5 and 1.291 against 0.328 in column 6. Every difference between the two blocs is significant, with p-values from 0.007 to below 0.001.

[Insert Table 9 about here]

Workers therefore persist through a failure when the work beneath it permits recovery and redirect their effort when it does not. Feature innovation improves performance both through the capabilities it supplies and through the recoverable disruption it generates.

Discussion

Continuous feature innovation creates an important tension for organizations that depend on digital platforms. New features are introduced to improve what workers can accomplish, yet introducing those features into systems already being used for ongoing work can also create instability and failure. A conventional reliability perspective suggests that these failures should erode at least some of the performance benefits created by innovation. Our findings reveal a more nuanced relationship. Greater use of newly released CRM features causally improves conversion performance while also increasing IT failure. More surprisingly, IT failure itself has a positive aggregate effect on conversion and transmits approximately 22% of the total effect of feature use on conversion. The direct and indirect pathways therefore complement rather than offset one another. In this setting, feature innovation creates value not only through the capabilities it provides, but also through how workers respond to some of the failures it generates.

Contributions to Research

Our findings contribute to research on digital innovation, IT reliability and failure, and the mechanisms through which technology use affects performance.

Continuous Feature Innovation: Capability and Disruption as Joint Pathways.

First, we extend research on digital innovation by showing that the performance consequences of continuous feature innovation cannot be understood solely through the capabilities that new features provide. Digital innovation research emphasizes the recombinability and continuing evolution of digital technologies (Nambisan et al., 2017; Yoo et al., 2010), while feature-level perspectives direct attention to how users engage particular technological capabilities within established systems (Karahanna et al., 2019). Our findings show that continuous feature innovation has another consequential property: new functionality is introduced into a production environment in which workers are already dependent on the system. Innovation can therefore simultaneously expand capability and generate operational instability.

This observation changes how the performance effects of continuous feature innovation can be conceptualized. A natural expectation is that new features create a positive capability pathway while any failures accompanying their introduction constitute an offsetting reliability cost. We find instead that the two pathways can be complementary. New CRM feature use directly increases performance, but it also increases IT failure, which in turn positively affects performance. The mediation analysis shows that this indirect pathway is economically meaningful, accounting for approximately 22% of the total effect of feature use on conversion.

The implication is not that instability is inherently valuable. Rather, our results suggest that capability creation and disruption can be jointly produced by continuous digital innovation, and both can become pathways through which innovation affects performance. This expands the theoretical treatment of continuous feature innovation beyond the question of whether newly introduced functionality is adopted and useful to include what happens when that functionality destabilizes the work system into which it is introduced.

IT Failure: From Reliability Loss to Recoverable Disruption.

Second, we contribute to research on IT reliability and failure by challenging the assumption that the performance consequence of failure is uniformly negative. Prior IS research provides strong reasons to regard reliability as a desirable property of information systems and failure as costly (Butler & Gray, 2006). At the organizational level, failures can destroy market value, require organizational responses, and interrupt technology-dependent operations (Bharadwaj et al., 2009; Triche & Walden, 2018). At the individual level, interruptions can impose attentional and performance costs (Addas & Pinsonneault, 2018). Our positive aggregate estimate for IT failure appears inconsistent with this conventional expectation.

The additional analyses suggest a way to reconcile these perspectives. What matters may not simply be whether technology fails, but whether the work interrupted by that failure remains recoverable. A failure during internal work can interrupt task execution without eliminating the underlying task opportunity. An agent whose attempt to record information fails can return to the record, reload the page, retry the action, and complete the work. The failure creates a disruption, but the worker retains an opportunity to respond to it. By contrast, a failure during customer-facing work can occur against an external clock. A failed communication with a waiting customer may compromise an interaction that cannot simply be recreated by retrying the same action later.

Our category-level decomposition is consistent with this distinction. Recording failures make the largest positive contribution to the aggregate failure effect, whereas failures involving customer-facing communication make the largest negative contribution. Because this decomposition is descriptive rather than causal, it should not by itself be interpreted as establishing recoverability as the causal moderator of failure's performance effect. However, the post-failure trace evidence provides an independent behavioral test of the proposed mechanism. Workers respond to internal-facing failures with greater retry, rapid reload, and faster resolution, whereas customer-facing failures are associated with rerouting, abandonment, and greater departure from routine. These behavioral differences are difficult to reconcile with a simple explanation in which highly engaged agents merely experience more failures and also perform better.

Together, these findings suggest a shift in the theoretical question surrounding IT failure. Rather than asking simply whether failure harms or improves performance, research should ask

when disrupted work permits recovery and how workers respond when it does. We introduce recoverability as a useful organizing concept for answering that question. Failure may become productive when it creates an interruption from which the worker can re-engage, adapt, and recover; it remains costly when the disruption compromises an opportunity that cannot readily be restored.

This perspective also connects IT reliability research to research on learning from failure. Sitkin (1992) argues that relatively small failures can generate learning when they are sufficiently bounded, permit rapid response, and occur in domains familiar enough to support interpretation. Digital work provides a setting in which such responses can be observed behaviorally. Retry, reload, recovery time, rerouting, and abandonment capture what workers actually do immediately after technology fails. The results suggest that the informational and adaptive consequences of failure depend importantly on the structure of the work in which the failure occurs.

Opening the Mechanism Between Technology Use and Performance.

Third, we contribute to research examining how technology use translates into measurable worker and organizational performance. Studies of technology-enabled work have established that technology use can improve individual productivity and performance, including in customer-facing work and, more recently, in settings involving generative AI (Ahearne et al., 2008; Brynjolfsson et al., 2025; Noy & Zhang, 2023). Yet an observed total performance effect does not reveal the mechanisms through which that value is produced. Our analysis opens this mechanism by decomposing the performance effect of feature use into a direct pathway and an indirect pathway operating through IT failure. This distinction matters because an analysis limited to the total effect would show that new CRM features improve conversion but would miss the counterintuitive process through which part of that improvement occurs. The causal mediation estimates show that the indirect effect through IT failure has the same sign as the direct effect: failure is a complementary mediator, not an offsetting cost.

More broadly, this finding illustrates why understanding the business value of digital innovation may require studying not only successful technology use but also the frictions generated during use. Some consequences conventionally classified as implementation or operational costs may interact with worker behavior in ways that alter the value ultimately realized from technology. The performance effects of digital innovation may therefore reside partly in workers' responses to imperfect technology, rather than solely in frictionless use of the capabilities that technology provides.

Implications for Practice

The findings also have implications for managers of continuously evolving digital platforms and for organizations that depend on those platforms for knowledge work. The most immediate implication is not that organizations should tolerate IT failure because failure improves performance. The aggregate positive relationship masks important differences across types of work. Some failures appear compatible with recovery and productive re-engagement, while others coincide with abandonment and lost opportunities. Treating all failures as equivalent can therefore obscure where reliability matters most.

This suggests a more selective approach to reliability management. Conventional operational metrics often emphasize aggregate failure rates, uptime, error incidence, or time lost to failure. These measures remain important, but they may be insufficient for prioritizing reliability investments. Managers should also consider where in the workflow failures occur, whether the interrupted task can be recovered, and what workers do after the failure.

Failures occurring during customer-facing activities deserve particular attention. When a failure disrupts communication with a customer, the worker may be unable to recreate the interaction simply by retrying the task. Such failures can therefore threaten opportunities that are perishable. Reliability investments in customer-facing workflows may generate value not merely by reducing downtime, but by protecting interactions that cannot easily be recovered once disrupted.

Internal workflows present a different managerial problem. Our findings do not imply that firms should intentionally introduce failures into these activities. Rather, they suggest that the consequences of failures that do occur may depend on whether systems make recovery easy. Platform designers can therefore complement investments in failure prevention with investments in failure recovery. Preserving entered information, enabling rapid retry, maintaining workflow state, providing intelligible error feedback, and allowing workers to resume interrupted tasks may reduce the cost of unavoidable failures and potentially facilitate productive re-engagement.

The results also have implications for the management of continuous software releases. Vendors commonly face pressure to increase release velocity while maintaining reliability. Our findings suggest that the relevant managerial tradeoff may be more granular than a simple choice between innovation speed and system stability. Release and testing resources could be prioritized according to the recoverability of the workflows affected by a feature. New functionality touching customer-facing or otherwise time-sensitive activities may warrant particularly stringent testing and staged deployment, whereas failures in highly recoverable internal workflows may pose different operational risks.

Real-user monitoring data can support this more selective approach. The same behavioral traces used in our analysis—retry, reload, recovery time, rerouting, abandonment, and workflow deviation—could supplement conventional technical reliability measures. A failure followed im-

mediately by successful retry conveys different operational information from an error followed by abandonment or termination of the session. Organizations can therefore evaluate failures not only by their technical incidence but also by their behavioral aftermath.

Taken together, these implications suggest a shift from pursuing reliability uniformly toward managing the consequences and recoverability of failure. Organizations should continue to prevent failures, particularly where failures threaten nonrecoverable customer or business opportunities. But when some failures are unavoidable in continuously evolving digital systems, managers can also design technologies and workflows that help workers recover effectively when disruption occurs.

Boundary Conditions and Future Research

Several features of our study define important boundaries for these conclusions and opportunities for future research. The empirical setting is a single SaaS CRM platform used by insurance agents, and the work is characterized by repeated interaction with a familiar digital environment. These characteristics may be especially conducive to recovery because workers can draw on accumulated task and system knowledge when failures occur. Whether similar responses emerge in less familiar, more safety-critical, or more tightly coupled work remains an open question.

The distinction between internal-facing and customer-facing failure also provides evidence consistent with recoverability rather than a direct causal estimate of recoverability itself. The Shapley decomposition is descriptive, and the post-failure analyses identify behavioral patterns consistent with the proposed mechanism. Future research could directly manipulate or exploit exogenous variation in task recoverability to identify its causal role. Such research could examine whether recovery opportunities, task familiarity, time pressure, failure severity, or the reversibility of interrupted work determine when failure becomes productive or costly.

Future research could also distinguish more explicitly among different sources of failure. In our setting, failures arise from both system-side and user-side causes, and their distribution varies across task categories. Understanding whether workers learn differently from failures they attribute to themselves versus failures they attribute to the technology could further refine the recoverability mechanism. Similarly, repeated exposure to failure may eventually generate frustration or workarounds rather than productive adaptation, suggesting that the benefits associated with bounded, recoverable disruption may have limits.

Finally, the continuing evolution of digital platforms makes the relationship between innovation and reliability increasingly important. As organizations adopt systems that are updated continuously—and as increasingly complex and probabilistic technologies become embedded in work—the assumption that innovation can first be implemented and then stabilized becomes less tenable. Future research should therefore examine innovation and reliability as jointly evolving properties of digital work systems rather than as separate stages of technology implementation.

References

Aanestad, M., & Jensen, T. B. (2016). Collective mindfulness in post-implementation IS adaptation processes. *Information and Organization*, *26*(1-2), 13–27. doi: 10.1016/j.infoandorg.2016.02.001

Adão, R., Kolesár, M., & Morales, E. (2019). Shift-share designs: Theory and inference. *The Quarterly Journal of Economics*, *134*(4), 1949–2010. doi: 10.1093/qje/qjz025

Addas, S., & Pinsonneault, A. (2015). The many faces of information technology interruptions: a taxonomy and preliminary investigation of their performance effects. *Information Systems Journal*, *25*(3), 231–273. doi: 10.1111/isj.12064

Addas, S., & Pinsonneault, A. (2018). E-Mail Interruptions and Individual Performance: Is There a Silver Lining? *MIS Quarterly*, *42*(2), 381–405. doi: 10.25300/MISQ/2018/13157

Ahearne, M., Jones, E., Rapp, A., & Mathieu, J. (2008). High touch through high tech: The impact of salesperson technology usage on sales performance via mediating mechanisms. *Management Science*, *54*(4), 671–685. doi: 10.1287/mnsc.1070.0783

Ahearne, M., Pourmasoudi, M., & Habel, J. (2025). Inside sales structures and firm performance. *Journal of the Academy of Marketing Science*, *53*(5), 1356–1378. doi: 10.1007/s11747-025-01085-6

Balachander, S., Ghosh, B., & Stock, A. (2010). Why bundle discounts can be a profitable alternative to competing on price promotions. *Marketing Science*, *29*(4), 624–638. doi: 10.1287/mksc.1090.0540

Banker, R., Liang, Y., & Ramasubbu, N. (2021). Technical Debt and Firm Performance. *Management Science*, *67*(5), 3174–3194. doi: 10.1287/mnsc.2019.3542

Basu, A. K., Lal, R., Srinivasan, V., & Staelin, R. (1985). Salesforce compensation plans: An agency theoretic perspective. *Marketing Science*, *4*(4), 267–291. doi: 10.1287/mksc.4.4.267

Beaudry, A., & Pinsonneault, A. (2005). Understanding User Responses to Information Technology: A Coping Model of User Adaptation1. *MIS Quarterly*, *29*(3), 493–524. doi: 10.2307/25148693

Benaroch, M., & Fink, L. (2021). No Rose without a thorn: Board IT competence and market reactions to operational IT failures. *Information & Management*, *58*(8), 103546. doi: 10.1016/j.im.2021.103546

Benbya, H., Nan, N., Tanriverdi, H., & Yoo, Y. (2020). Complexity and Information Systems Research in the Emerging Digital World. *MIS Quarterly*, *44*(1), 1–18. doi: 10.25300/MISQ/2020/13304

Bharadwaj, A., Keil, M., & Mähring, M. (2009). Effects of Information Technology Failures on the Market Value of Firms. *Journal of Strategic Information Systems*, *18*(2), 66–79. doi: 10.1016/j.jsis.2009.04.001

Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-Experimental Shift-Share Research Designs. *The Review of Economic Studies*, *89*(1), 181–213. doi: 10.1093/restud/rdab030

Borusyak, K., Hull, P., & Jaravel, X. (2025). A Practical Guide to Shift-Share Instruments. *Journal of Economic Perspectives*, *39*(1), 181–204. doi: 10.1257/jep.20231370

Brynjolfsson, E., Li, D., & Raymond, L. (2025). Generative AI at Work. *The Quarterly Journal of Economics*, *140*(2), 889–942. doi: 10.1093/qje/qjae044

Butkiewicz, M., Madhyastha, H. V., & Sekar, V. (2011). Understanding website complexity: measurements, metrics, and implications. In *Proceedings of the 2011 ACM SIGCOMM conference on Internet measurement conference* (pp. 313–328). New York, NY, USA: Association for Computing Machinery. doi: 10.1145/2068816.2068846

Butler, B. S., & Gray, P. H. (2006). Reliability, Mindfulness, and Information Systems. *MIS Quarterly*, *30*(2), 211–224. doi: 10.2307/25148728

Celli, V. (2022). Causal Mediation Analysis in Economics: Objectives, Assumptions, Models. *Journal of Economic Surveys*, *36*(1), 214–234. doi: 10.1111/joes.12452

Chung, D. J., & Narayandas, D. (2017). Incentives versus reciprocity: Insights from a field experiment. *Journal of Marketing Research*, *54*(4), 511–524. doi: 10.1509/jmr.15.0174

Cito, J., Gotowka, D., Leitner, P., Pelette, R., Suljoti, D., & Dustdar, S. (2015). Identifying web performance degradations through synthetic and real-user monitoring. *Journal of Web Engineering*, *14*(5-6), 414–442.

Davis, G. B., Lee, A. S., Nickles, K. R., Chatterjee, S., Hartung, R., & Wu, Y. (1992). Diagnosis of an information system failure. *Information & Management*, *23*(5), 293–318. doi: 10.1016/0378-7206(92)90059-o

Demirer, M., Horton, J. J., Immorlica, N., Lucier, B., & Shahidi, P. (2026). *Chaining Tasks, Redefining Work: A Theory of AI Automation* (Working Paper Series No. 34859). National Bureau of Economic Research. Retrieved from https://www.nber.org/papers/w34859 doi: 10.3386/w34859

Dippel, C., Ferrara, A., & Heblich, S. (2020). Causal Mediation Analysis in Instrumental-Variables Regressions. *The Stata Journal*, *20*(3), 613–626. doi: 10.1177/1536867X20953572

Elie-Dit-Cosaque, C. M., & Straub, D. W. (2011). Opening the black box of system usage: user adaptation to disruptive IT. *European Journal of Information Systems*, *20*(5), 589–607. doi: 10.1057/ejis.2010.23

Feng, Y. K., Yu, N. Y., Tam, K. Y., & Lai, M. C. (2024). Effective Intervention in User Adaptation: A 2×2 Coping Framework Entailing Implementation and Contextual Factors. *Journal of the Association for Information Systems*, *25*(4), 962–989. doi: 10.17705/1jais.00884

Fong, J., & Waisman, C. (2025). The effects of delay in bargaining: Evidence from eBay. *Management Science*, *71*(12), 9976–9997. doi: 10.1287/mnsc.2023.01562

Fritz, M. S., & MacKinnon, D. P. (2007). Required Sample Size to Detect the Mediated Effect. *Psychological Science*, *18*(3), 233–239. doi: 10.1111/j.1467-9280.2007.01882.x

Frölich, M., & Huber, M. (2017). Direct and Indirect Treatment Effects: Causal Chains and Mediation Analysis with Instrumental Variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *79*(5), 1645–1666. doi: 10.1111/rssb.12232

Galluch, P., Grover, V., & Thatcher, J. (2015). Interrupting the Workplace: Examining Stressors in an Information Technology Context. *Journal of the Association for Information Systems*, *16*(1), 1–47. doi: 10.17705/1jais.00387

Gans, J. S., & Goldfarb, A. (2026). *O-Ring Automation* (Working Paper Series No. 34639). National Bureau of Economic Research. Retrieved from https://www.nber.org/papers/w34639 doi: 10.3386/w34639

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, *110*(8), 2586–2624. doi: 10.1257/aer.20181047

Grover, V., Lindberg, A., Benbasat, I., & Lyytinen, K. (2020). The perils and promises of big data research in information systems. *Journal of the Association for Information Systems*, *21*(2), 268–291. doi: 10.17705/1jais.00601

Guesalaga, R., & Kapelianis, D. (2015). When do salespeople pursue and win deals? A two-stage model of sales opportunity outcomes. *Journal of Business and Industrial Marketing*, *30*(7), 817–829. doi: 10.1108/JBIM-06-2014-0120

Hartmann, N. N., Wieland, H., Chaker, N. N., Habel, J., & Tyler, F. J. (2025). Sales content platforms and opportunity conversion. *Production and Operations Management*, 10591478251413339. doi: 10.1177/10591478251413339

Imai, K., Keele, L., & Tingley, D. (2010). A General Approach to Causal Mediation Analysis. *Psychological Methods*, *15*(4), 309–334. doi: 10.1037/a0020761

Karahanna, E., Chen, A., Liu, Q. B., & Serrano, C. (2019). Capitalizing on Health Information Technology to Enable Digital Advantage in U.S. Hospitals. *MIS Quarterly*, *43*(1), 113–140. doi: 10.25300/

MISQ/2019/12743
Keramidis, P., & Shollo, A. (2026). The realisation of IS business value: a value-reflexive perspective. *European Journal of Information Systems*, *35*(2), 280–300. doi: 10.1080/0960085x.2025.2607449
Kremer, M. (1993). The O-Ring Theory of Economic Development. *The Quarterly Journal of Economics*, *108*(3), 551–575. doi: 10.2307/2118400
Lindberg, A. (2020). Developing theory through integrating human and machine pattern recognition. *Journal of the Association for Information Systems*, *21*(1), 90–116. doi: 10.17705/1jais.00593
MacKinnon, D. P., Lockwood, C. M., & Williams, J. (2004). Confidence Limits for the Indirect Effect: Distribution of the Product and Resampling Methods. *Multivariate Behavioral Research*, *39*(1), 99–128. doi: 10.1207/s15327906mbr3901_4
Mattern, J., Tarafdar, M., Klein, S., & Schellhammer, S. (2024). Thriving in a bruising job: How high achieving IT professionals can cope with occupational demands. *Information Systems Journal*, *34*(6), 1902–1934. doi: 10.1111/isj.12513
Najjar, M. S., Kettinger, W. J., & Kettinger, L. D. (2022). IS incident recovery and service value: a service-dominant logic view. *European Journal of Information Systems*, *31*(4), 492–524. doi: 10.1080/0960085x.2020.1869915
Nambisan, S., Lyytinen, K., Majchrzak, A., & Song, M. (2017). Digital Innovation Management: Reinventing Innovation Management Research in a Digital World. *MIS Quarterly*, *41*(1), 223–238. doi: 10.25300/MISQ/2017/41:1.03
Nan, N., & Tanriverdi, H. (2017). Unifying the role of IT in hyperturbulence and competitive advantage via a multilevel perspective of IS strategy. *MIS Quarterly*, *41*(3), 937–A8. doi: 10.25300/MISQ/2017/41.3.12
Nastjuk, I., Trang, S., Grummeck-Braamt, J.-V., Adam, M. T. P., & Tarafdar, M. (2024). Integrating and synthesising technostress research: a meta-analysis on technostress creators, outcomes, and IS usage contexts. *European Journal of Information Systems*, *33*(3), 361–382. doi: 10.1080/0960085x.2022.2154712
Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, *381*(6654), 187–192. doi: 10.1126/science.adh2586
Pirkkalainen, H., Salo, M., Tarafdar, M., & Makkonen, M. (2019). Deliberate or Instinctive? Proactive and Reactive Coping for Technostress. *Journal of Management Information Systems*, *36*(4), 1179–1212. doi: 10.1080/07421222.2019.1661092
Poláková-Kersten, M., Khanagha, S., van den Hooff, B., & Khapova, S. N. (2023). Digital transformation in high-reliability organizations: A longitudinal study of the micro-foundations of failure. *The Journal of Strategic Information Systems*, *32*(1), 101756. doi: 10.1016/j.jsis.2023.101756
Ragu-Nathan, T. S., Tarafdar, M., Ragu-Nathan, B. S., & Tu, Q. (2008). The Consequences of Technostress for End Users in Organizations: Conceptual Development and Empirical Validation. *Information Systems Research*, *19*(4), 417–433. doi: 10.1287/isre.1070.0165
Rao, R. S., Viswanathan, M., John, G., & Kishore, S. (2021). Do activity-based incentive plans work? Evidence from a large-scale field intervention. *Journal of Marketing Research*, *58*(4), 686–704. doi: 10.1177/00222437211020013
Rinta-Kahila, T., Penttinen, E., Salovaara, A., Soliman, W., & Ruissalo, J. (2023). The Vicious Circles of Skill Erosion: A Case Study of Cognitive Automation. *Journal of the Association for Information Systems*, *24*(5), 1378–1412. doi: 10.17705/1jais.00829
Sabnis, G., Chatterjee, S. C., Grewal, R., & Lilien, G. L. (2013). The sales lead black hole: On sales reps' follow-up of marketing leads. *Journal of Marketing*, *77*(1), 52–67. doi: 10.1509/jm.10.0047
Schmitz, C., Lee, Y.-C., & Lilien, G. L. (2014). Cross-selling performance in complex selling contexts: An examination of supervisory- and compensation-based controls. *Journal of Marketing*, *78*(3), 1–19.

doi: 10.1509/jm.12.0421

Schryen, G. (2013). Revisiting IS business value research: what we already know, what we still need to know, and how we can get there. *European Journal of Information Systems*, *22*(2), 139–169. doi: 10.1057/ejis.2012.45

Simon, H. A. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society*, *106*(6), 467–482.

Sitkin, S. B. (1992). Learning Through Failure: The Strategy of Small Losses. *Research in Organizational Behavior*, *14*, 231–266.

Sobel, M. E. (1982). Asymptotic Confidence Intervals for Indirect Effects in Structural Equation Models. *Sociological Methodology*, *13*, 290–312. doi: 10.2307/270723

Spetz, J., & Maiuro, L. S. (2004). Measuring Levels of Technology in Hospitals. *The Quarterly Review of Economics and Finance*, *44*(3), 430–447. doi: 10.1016/j.qref.2004.05.006

Tan, T. F., & Netessine, S. (2014). When does the devil make work? An empirical study of the impact of workload on worker productivity. *Management Science*, *60*(6), 1574–1593. doi: 10.1287/mnsc.2014.1950

Tanriverdi, H., & Du, K. (2020). Corporate strategy changes and information technology control effectiveness in multibusiness firms. *MIS Quarterly*, *44*(4), 1573–1617. doi: 10.25300/MISQ/2020/14223

Tanriverdi, H., Kwon, J., & Im, G. (2025). Taming Complexity in the Cybersecurity of Multihospital Systems: The Role of Enterprise-wide Data Analytics Platforms. *MIS Quarterly*, *49*(1), 243–273. doi: 10.25300/MISQ/2024/17752

Tarafdar, M., Pullins, E. B., & Ragu-Nathan, T. S. (2015). Technostress: negative effect on performance and possible mitigations. *Information Systems Journal*, *25*(2), 103–132. doi: 10.1111/isj.12042

Tiwana, A., & Safadi, H. (2024). Atrophy in Aging Systems: Evidence, Dynamics, and Antidote. *Information Systems Research*, *35*(1), 66–86. doi: 10.1287/isre.2023.1218

Tiwana, A., & Safadi, H. (2026). Silence Inside Systems: Roots and Generativity Consequences. *Information Systems Research*, *37*(1), 584–602. doi: 10.1287/isre.2022.0586

Triche, J. H., & Walden, E. (2018). The Use of Impression Management Strategies to Manage Stock Market Reactions to IT Failures. *Journal of the Association for Information Systems*, *19*(4), 333–357. doi: 10.17705/1jais.00494

Vieira da Cunha, J., Carugati, A., & Leclercq-Vandelannoitte, A. (2015). The dark side of computer-mediated control. *Information Systems Journal*, *25*(4), 319–354. doi: 10.1111/isj.12066

Wang, X. S., Balasubramanian, A., Krishnamurthy, A., & Wetherall, D. (2013). Demystifying page load performance with WProf. In *Proceedings of the 10th USENIX conference on Networked Systems Design and Implementation* (pp. 473–486). USA: USENIX Association.

Weinert, C., Maier, C., Laumer, S., & Weitzel, T. (2022). Repeated IT Interruption: Habituation and Sensitization of User Responses. *Journal of Management Information Systems*, *39*(1), 187–217. doi: 10.1080/07421222.2021.2023411

Xu, Y., Tan, T. F., & Netessine, S. (2022). The impact of workload on operational risk: Evidence from a commercial bank. *Management Science (INFORMS)*, *68*(4), 2668–2693. doi: 10.1287/mnsc.2021.4019

Yoo, Y., Henfridsson, O., & Lyytinen, K. (2010). The New Organizing Logic of Digital Innovation: An Agenda for Information Systems Research. *Information Systems Research*, *21*(4), 724–735. doi: 10.1287/isre.1100.0322

Zhao, X., Xia, Q., & Huang, W. (2020). Impact of technostress on productivity from the theoretical perspective of appraisal and coping processes. *Information & Management*, *57*(8), 103265. doi: 10.1016/j.im.2020.103265

Table 1. Prior Studies of IT Failure, Interruption and Worker Response

Study	Construct examined	Unit	Mechanism
IT failure and firm value			
Bharadwaj et al. (2009)	IT failures and market value loss	Firm	Event study within a resource-based view
Benaroch and Fink (2021)	Board IT competence moderating market reaction to operational failures	Firm	Agency theory, governance reduces investor uncertainty
Najjar et al. (2022)	IS incident recovery and service value	Organization	Service-dominant logic, value co-created during recovery
ICT interruptions and performance			
Addas and Pinsonneault (2015)	Interruption taxonomy and individual performance	Individual	Attentional depletion and task-switching cost
Galluch et al. (2015)	ICT interruptions, stressors and strain	Individual	Stressor-strain model
Weinert et al. (2022)	Repeated interruption, habituation and sensitization	Individual	Conditioning theory, non-linear exposure response
Technostress			
Ragu-Nathan et al. (2008)	Technostress creators, job satisfaction and commitment	Individual	Transactional stress model with organizational inhibitors
Tarafdar et al. (2015)	Technostress and technology-enabled performance	Individual	Role theory and sociotechnical theory
Zhao et al. (2020)	Application-level stressors, appraisal, coping and productivity	Individual	Transactional theory of stress at the application level
Nastjuk et al. (2024)	Meta-analysis of technostress creators and outcomes	Individual	Synthesis of 102 studies
User coping and adaptation			
Beaudry and Pinsonneault (2005)	Four adaptation strategies from primary and secondary appraisal	Individual	Coping theory applied to IT events
Elie-Dit-Cosaque and Straub (2011)	Measures of system usage and user adaptation to disruptive IT	Individual	Coping model of user adaptation, tested in a two-by-two laboratory experiment
Pirkkalainen et al. (2019)	Proactive and reactive coping moderating technostress	Individual	Proactive versus reactive coping
Feng et al. (2024)	Intervention type by implementation phase and voluntariness	Individual	Coping extended with a time dimension
Mattern et al. (2024)	Coping portfolios and strain levels	Individual	Requisite variety and coping repertoire breadth

continued on the next page

Table 1. Prior Studies of IT Failure, Interruption and Worker Response (continued)

Study	Construct examined	Unit	Mechanism
System atrophy			
Tiwana and Safadi (2024)	System atrophy, bugginess and developer disengagement	System	Modularity dynamics
Tiwana and Safadi (2026)	System silence and generativity	System	Attention funnelling
Banker et al. (2021)	Technical debt in CRM systems and firm performance	Firm	Shortcuts compounding maintenance obligations
High-reliability organizing			
Butler and Gray (2006)	Reliable performance through mindfulness	Organization and individual	Routine-based versus mindfulness-based reliability
Aanestad and Jensen (2016)	Collective mindfulness in post-implementation adaptation	Organization	High-reliability organizing applied after implementation
Poláková-Kersten et al. (2023)	Tension between high-reliability identity and digital transformation	Organization	Identity threat producing self-protective behaviour in the IT workforce
Rinta-Kahila et al. (2023)	Skill erosion, dependency and vulnerability	Individual and organization	Vicious circle in which automation removes practice
IS business value			
Schryen (2013)	Systematic review of IS business value research	Firm	Bibliometric synthesis
Keramidis and Shollo (2026)	Value realization as enacted practice	Organization	Value-reflexive perspective
CRM control and the representation of work			
Vieira da Cunha et al. (2015)	Burden of representation under loose coupling	Individual and organization	Loose versus tight coupling of work and record

Table 2. Variables Description

Variable	Definition and construction	
Technology use.		
	Feature-level input	**Agent-week construction**
New CRM Feature Use	A Saidin novelty-weighted index of an agent's engagement with newly released CRM features (Karahanna et al., 2019; Spetz & Maiuro, 2004). u_{kit} is agent i's weekly click count on feature k, and A_{kit} indicates whether agent i has adopted feature k by week t. Rare features receive high weight and widely adopted features near-zero weight; non-adopters contribute zero by construction. The feature set is itself time-varying, expanding with each platform-controlled release wave, and the weights are recomputed each week to reflect both the growing set and the evolving cross-agent adoption distribution.	$a_{kt} = 1 - \frac{1}{N}\sum_i A_{kit}$ $New\,CRM\,Feature\,Use_{it} = \sum_k a_{kt}\,u_{kit}$
New GenAI Use	A Saidin index on the same novelty-weighting logic, applied to GenAI releases. GenAI reaches agents as capability updates: each release updates the platform's GenAI functionality and supersedes the one before it, so a single version is active in any given week. G_{it} is agent i's GenAI minutes in week t, $k(t)$ the version active in week t, and r_k the release week of version k. Adoption accumulates within a version and resets when the next begins, so the weight tracks the novelty of whichever version is currently active. Releases falling in the same week form a single version, effective from the next complete week.	$G_{ikt} = G_{it}\,\mathbf{1}\{k = k(t)\}$ $A_{ikt} = \mathbf{1}\Big\{\sum_{s=r_k}^{t} G_{iks} > 0\Big\}$ $a_{kt} = 1 - \frac{1}{N}\sum_i A_{ikt}$ $New\,GenAI\,Use_{it} = \sum_k a_{kt} G_{ikt}$ $= a_{k(t),t}\,G_{it}$
Feature Breadth	The share of distinct platform workflow features accessed by agent i in week t — unique features visited, divided by the total distinct feature groups observed across all panel agents.	
Platform Time	The total time agent i spends in the platform in week t, summed over the week's sessions: $\sum_{s\in S_{it}} d_{is}$, where d_{is} is session s's total duration and S_{it} is the set of agent i's sessions in week t.	
	Session-level input	**Agent-week construction**
IT Failure	Within session s, F_{isk} is the set of tasks of category k that fail, d_j the time the agent spends on task j, and d_{is} the session's total duration. A task is recorded as failed when an error is logged against it, so the time spent on it yields no completed work. S_{it} is the set of agent i's sessions in week t. IT Failure enters estimation on the agent-weeks in which failure occurs, with an incidence dummy included as a control.	$\tau_{isk} = \frac{\sum_{j\in F_{isk}} d_j}{d_{is}}$ $IT\,Failure_{ikt} = \frac{1}{\lvert S_{it}\rvert}\sum_{s\in S_{it}} \tau_{isk}$ $IT\,Failure_{it} = \sum_{k=1}^{5} IT\,Failure_{ikt}$
Incidence of Failure	An indicator equal to one if agent i records at least one failed task in week t, and zero otherwise.	$D_{it} = \mathbf{1}\Big\{\sum_{s\in S_{it}}\sum_k \lvert F_{isk}\rvert > 0\Big\}$

continued on the next page

Table 2. Variables Description (continued)

Variable	Definition and construction

Performance. Sales performance is captured using conversion, the rate at which active later-stage deals resolve as successful policy purchases in a given agent-week, reflecting precision-oriented execution effectiveness at the point where deals close.

	Deal-level input	Agent-week construction
Conversions	Let $\mathcal{D}_{it}$ denote the set of active deals for agent i in week t across both stages, and let N_{it} denote the number of later-stage deals that convert in week t.	$= \dfrac{N_{it}}{\lvert\mathcal{D}_{it}\rvert}$

Portfolio characteristics. Each of these controls summarises the agent's active deal portfolio in the week as an average over the deals it contains. Let $\mathcal{D}_{it}$ denote agent i's set of active deals in week t. Because the agent manages early-stage and later-stage deals as a coupled system, the components carried at one stage remain a concurrent condition for performance at the other. Both complicatedness dimensions therefore enter at both stages.

	Deal-level input	Agent-week construction	Theory
Product Complicatedness	p_d, the product category associated with deal d.	$= \lvert\{\, p_d \,:\, d \in \mathcal{D}_{it}^{s} \,\}\rvert$	Complicatedness captures the breadth of distinct components in an agent's deal portfolio, separated from complexity, which emerges through the form and dynamics of the interactions among those components (Benbya et al., 2020; Tanriverdi & Du, 2020; Tanriverdi et al., 2025). Let $s \in \{Q,\ C\}$ index the stage of the funnel, where Q denotes qualification-stage deals and C denotes conversion-stage deals; both dimensions are constructed at either stage from the same stage-specific deal set.
Customer Complicatedness	c_d, the customer segment associated with deal d.	$= \lvert\{\, c_d \,:\, d \in \mathcal{D}_{it}^{s} \,\}\rvert$	
Process activity burden	A_d, the cumulative number of transactions and activities recorded for deal d as it progresses through the CRM workflow.	$= \dfrac{\sum_{d \in \mathcal{D}_{it}} A_d}{\lvert\mathcal{D}_{it}\rvert}$	The empirical models also control for the average accumulated transaction and activity burden embedded in an agent's active deal portfolio.
Bundling intensity	B_d, the number of products linked to deal d.	$= \dfrac{\sum_{d \in \mathcal{D}_{it}} B_d}{\lvert\mathcal{D}_{it}\rvert}$	Cross-selling and promotional-bundling behavior is an established, distinct driver of conversion performance in the sales literature (Balachander et al., 2010; Schmitz et al., 2014). Bundling intensity proxies for this behavior.

continued on the next page

Table 2. Variables Description (continued)

Variable	Definition and construction		
Portfolio Value	V_d, the raw dollar value of deal d.	$= \frac{\sum_{d \in \mathcal{D}_{it}} V_d}{\lvert \mathcal{D}_{it} \rvert}$	Financial incentive structure is an established, direct driver of sales effort and performance in the compensation literature (Basu et al., 1985; Chung & Narayandas, 2017; Rao et al., 2021). Because compensation in this setting is commission-linked, portfolio value proxies for this incentive: higher-value portfolios imply higher potential earnings.
Portfolio Age	G_{dt}, the number of days elapsed since deal d's creation date, as of week t.	$= \frac{\sum_{d \in \mathcal{D}_{it}} G_{dt}}{\lvert \mathcal{D}_{it} \rvert}$	Deal-cycle velocity, the idea that deals remaining open longer are subject to different urgency and closing dynamics than recently opened ones, is an established construct in sales performance research (Fong & Waisman, 2025; Guesalaga & Kapelianis, 2015). Portfolio age proxies for this dynamic.
Portfolio Size	Deal d is active in week t.	$= \lvert \mathcal{D}_{it} \rvert$	Workload, the volume of concurrent work an individual manages, is an established driver of individual performance, both generally in the operations literature (Tan & Netessine, 2014; Xu et al., 2022) and specifically for sales conversion in the marketing literature on deal volume (Ahearne et al., 2025; Sabnis et al., 2013). Portfolio size proxies for this constraint.

Table 3. Where IT Failure Falls

Task Category	Share of incidence (%)	Share of failure time (%)	Agent-weeks affected (%)
(1)	(2)	(3)	(4)
Communicating	47.1	41.3	24.1
Recording	35.0	38.2	16.0
Integrating	11.5	15.4	11.4
Coordinating	4.4	2.6	4.3
Curating	2.1	2.5	2.4
Total	100.0	100.0	

Note. Failures total 2.67% of session time. Columns 2 and 3 are shares of that total on two bases: incidence counts disrupted task, failure time counts share of session time lost. Column 4 is the share of agent-weeks in which the group appears at all.

Table 4. Source of IT Failure

Task Category	System-caused (%)	User-caused (%)	Group total (%)
(1)	(2)	(3)	(4)
Communicating	30.9	10.4	41.3
Recording	0.5	37.7	38.2
Integrating	8.6	6.9	15.4
Coordinating	0.8	1.8	2.6
Curating	1.1	1.4	2.5
Total	41.9	58.2	100.0

Note. All entries are shares of total failure time. Columns 2 and 3 add across to column 4, which reproduces column 3 of Table 2. A category's internal composition is column 3 divided by column 4. Row and column totals may differ from their components by 0.1 owing to rounding.

Table 5. Mediator Equation: IT Failure

	IT Failure	
	(1)	(2)
New CRM Feature Use	**-0.022****	**0.248*****
	(0.008)	(0.074)
New GenAI Use	-0.016	-0.018
	(0.011)	(0.014)
Portfolio Size	-0.018	-0.019
	(0.024)	(0.027)
Portfolio Age	0.034**	0.034**
	(0.014)	(0.015)
Portfolio Value	-0.020	-0.028*
	(0.015)	(0.017)
Bundling intensity	0.003	0.014
	(0.022)	(0.024)
Process activity burden	-0.006	-0.009
	(0.007)	(0.007)
Product Complicatedness (Conversion-Stage)	-0.003	-0.009
	(0.017)	(0.018)
Customer Complicatedness (Conversion-Stage)	0.007	0.006
	(0.015)	(0.016)
Product Complicatedness (Qualification-Stage)	0.010	-0.011
	(0.023)	(0.029)
Customer Complicatedness (Qualification-Stage)	-0.017	-0.022
	(0.025)	(0.026)
Feature Breadth	-0.031***	-0.130***
	(0.009)	(0.029)
Platform Time	0.045***	-0.000
	(0.012)	(0.017)
Constant	-0.004	—
	(0.003)	
Estimator	OLS	2SLS
Observations	16,408	16,408
Agent FE	Yes	Yes
Week FE	Yes	Yes
Adj R-sq	0.172	-0.131
Model F-statistic	4.001	3.257
Kleibergen–Paap rk LM	—	62.980***
Cragg-Donald Wald F	—	159.400***

Note. Clustered standard errors. *** p<0.01, ** p<0.05, * p<0.1. Column 2 instruments New CRM Feature Use.

Table 6. Outcome Equation: Conversion

	Conversion				
	(1)	(2)	(3)	(4)	(5)
IT Failure			**0.023****	**0.034*****	**0.266****
			(0.011)	(0.012)	(0.118)
New CRM Feature Use		**0.018****	**0.019****	**0.291*****	**0.235*****
		(0.009)	(0.009)	(0.088)	(0.085)
Incidence of Failure	0.025	0.026	0.023	0.025	0.001
	(0.018)	(0.018)	(0.017)	(0.018)	(0.022)
New GenAI Use	0.013	0.013	0.013	0.011	0.015
	(0.009)	(0.010)	(0.009)	(0.012)	(0.011)
Portfolio Size	0.387***	0.387***	0.387***	0.386***	0.391***
	(0.044)	(0.044)	(0.044)	(0.043)	(0.043)
Portfolio Age	-0.442***	-0.441***	-0.442***	-0.442***	-0.451***
	(0.031)	(0.031)	(0.031)	(0.031)	(0.031)
Portfolio Value	0.218***	0.217***	0.218***	0.210***	0.217***
	(0.029)	(0.029)	(0.029)	(0.031)	(0.030)
Bundling intensity	0.136***	0.137***	0.137***	0.148***	0.146***
	(0.038)	(0.038)	(0.038)	(0.038)	(0.036)
Process activity burden	0.236***	0.236***	0.236***	0.234***	0.235***
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Product Complicatedness (Conversion-Stage)	0.448***	0.447***	0.447***	0.440***	0.442***
	(0.035)	(0.035)	(0.035)	(0.036)	(0.035)
Customer Complicatedness (Conversion-Stage)	0.151***	0.151***	0.151***	0.149***	0.148***
	(0.029)	(0.029)	(0.029)	(0.030)	(0.029)
Product Complicatedness (Qualification-Stage)	0.021	0.019	0.019	-0.002	-0.000
	(0.036)	(0.036)	(0.036)	(0.037)	(0.036)
Customer Complicatedness (Qualification-Stage)	0.037	0.037	0.037	0.032	0.037
	(0.049)	(0.049)	(0.049)	(0.046)	(0.045)
Feature Breadth	-0.030***	-0.037***	-0.036***	-0.136***	-0.105***
	(0.011)	(0.012)	(0.012)	(0.035)	(0.035)
Platform Time	0.027*	0.024	0.023	-0.023	-0.017
	(0.015)	(0.015)	(0.015)	(0.021)	(0.021)
Constant	0.737***	0.737***	0.738***	—	—
	(0.008)	(0.008)	(0.008)		
Estimator	OLS	OLS	OLS	2SLS	2SLS
Observations	16,408	16,408	16,408	16,408	16,408
Agent FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Adj R-sq	0.674	0.675	0.675	0.452	0.450
Model F-statistic	195.122	181.380	170.082	163.104	169.381
Kleibergen–Paap rk LM	—	—	—	63.532***	79.970***
Cragg-Donald Wald F	—	—	—	162.932***	65.936***

Note. Clustered standard errors. *** p<0.01, ** p<0.05, * p<0.1. Column 4 instruments New CRM Feature Use alone; column 5 instruments New CRM Feature Use and IT Failure.

Table 7. First-Stage Instrument Relevance

	New CRM Feature Use		*IT Failure*
	(1)	(2)	(3)
Z_{it}^{feature}	-0.721***	-0.720***	-0.179***
	(0.087)	(0.087)	(0.046)
Z_{it}^{failure}	—	-0.044***	0.111***
		(0.016)	(0.012)
Observations	16,408	16,408	16,408
Controls	Yes	Yes	Yes
Agent/Week FE	Yes/Yes	Yes/Yes	Yes/Yes
R-squared	0.478	0.478	0.213
SW F-statistic	68.30	64.38	88.22
KP rk LM	62.98	79.97	79.97
KP Wald rk F	68.30	40.40	40.40
CD Wald F	159.40	65.94	65.94

Note. Standard errors clustered by agent in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Column 1 is the first stage of the single-endogenous system estimated in column 2 of Table 5; columns 2 and 3 are the two first stages of the system estimated in column 5 of Table 6.

Table 8. Causal Mediation Results

	Estimate	BC 95% CI
a-path: *New CRM Feature Use* → *IT Failure*	0.248***	[0.126, 0.419]
b-path: *IT Failure* → *Conversion*	0.266**	[0.023, 0.509]
Average causal mediation effect (ACME)	0.066**	[0.006, 0.152]
Average direct effect (ADE)	0.235***	[0.054, 0.402]
Average total effect (ATE)	0.300***	[0.137, 0.493]

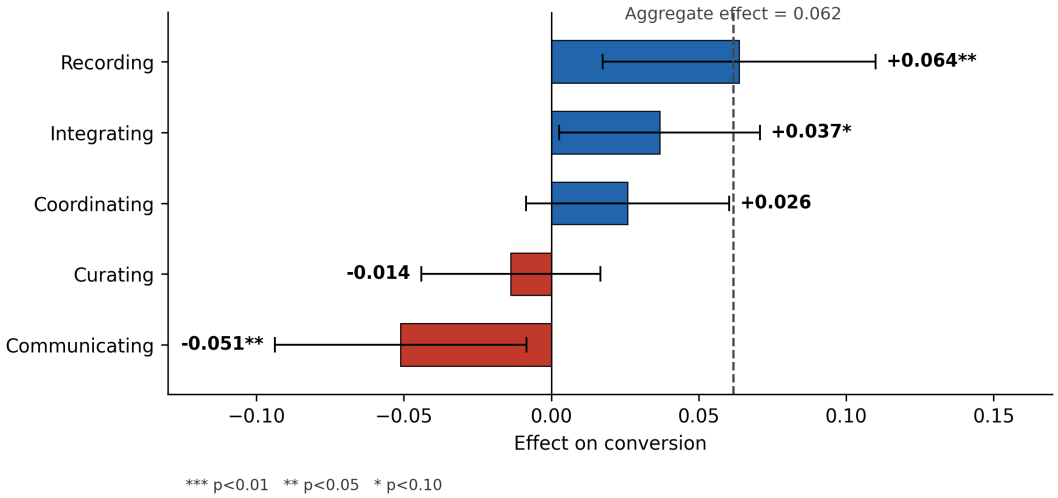
Note. *** p<0.01, ** p<0.05, * p<0.10. Intervals are bias-corrected (BC) bootstrap percentile intervals, which shift the percentile endpoints to correct for the skew in the bootstrap distribution of a product of coefficients.

Table 9. IT Failure and Post-Failure Behavior

	Page retry	**Short reload**	**Recovery time**	**Reroute rate**	**Abandonment**	**Workflow deviation**
	(1)	(2)	(3)	(4)	(5)	(6)
Internal-facing failure	0.069***	0.106***	−0.319***	0.019	0.549***	0.328***
	(0.021)	(0.022)	(0.039)	(0.039)	(0.050)	(0.048)
Customer-facing failure	−0.025	0.003	−0.132***	0.659***	0.916***	1.291***
	(0.026)	(0.026)	(0.049)	(0.057)	(0.067)	(0.063)
Difference	**0.094*** **	**0.103*** **	**−0.187*** **	**−0.640*** **	**−0.367*** **	**−0.964*** **
p (difference)	0.007	0.004	0.001	<0.001	<0.001	<0.001
Observations (agents)	12,094	12,094	6,489	12,094	12,094	12,094
	(571)	(571)	(529)	(571)	(571)	(571)

Note. Each column is a separate OLS regression of a standardized post-failure behaviour on the two failure-bloc measures. Internal-facing pools coordinating, recording and integrating; customer-facing pools curating and communicating. Columns 1–2 measure persistence, columns 3–6 disengagement or delay. Difference is the internal-facing coefficient minus the customer-facing coefficient, both estimated in the same regression, and its p-value is a Wald test of their equality. Column 3 is estimated only on agent-weeks in which a failure occurred. *** p<0.01, ** p<0.05, * p<0.10.

Figure 1. Shapley Decomposition of the Aggregate IT-Failure Effect



Note. The figure decomposes the aggregate effect of IT failure on conversion into the contribution of each of five categories of knowledge work. A category's contribution is its average marginal contribution across all possible combinations of the five categories, so the contributions sum to the aggregate. The aggregate effect of 0.062 is the estimate obtained when all five categories enter together. Each estimate is from OLS on conversion with the same controls and fixed effects as the main specifications. Standard errors are bootstrapped and clustered by agent.